Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
9.	(a)			**Indicative content** 1. Oxidation states in Group 4 are stable at +4 at the top of the group and +2 at the bottom 2. Due to the inert pair effect 3. The inert pair effect increases down the group 4. PbO more stable than PbO_2 / PbO_2 is an oxidising agent / CO_2 more stable than CO / CO is a reducing agent 5. SiO doesn't form as this is a +2 oxidation state which cannot form this high in the group 6. +5 oxidation state possible in third period of Group 5 but not in Period 2 7. This is because phosphorus can expand its octet / phosphorus has available d-orbitals 8. NCl_5 not possible as N has no available d-orbitals so it cannot expand its octet 9. One characteristic of transition metals is that they form a variety of oxidation states 10. Mn can form a range of oxidation states as the 3d orbitals (and 4s) have similar energies and similar ionisation energies **5-6 marks:** At least seven relevant points, discussing the oxidation states of all three sets of elements *The candidate constructs a relevant, coherent and logically structured account including all key elements of the indicative content, clearly showing an understanding of the reasons for the differing oxidation states in all three categories. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.*	6	4	2			

Question				Marking details	Total	AO1	AO2	AO3	Maths	Prac
				3-4 marks: At least five relevant points, discussing the oxidation states of two sets of elements *The candidate constructs a coherent account including most of the key elements of the indicative content and little irrelevant material, explaining at least one of the inert pair effect or octet expansion. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.* **1-2 marks:** Two or three relevant points, discussing the oxidation states of any of the elements *The candidate attempts to link at least three relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material, however there is some understanding of the patterns in oxidation states. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks:** *The candidate does not make any attempt or give an answer worthy of credit.*						
	(b)	(i)		rate = k $[Pt^{4+}]$ $[Sn^{2+}]$	1		1			
		(ii)		(rate constant is large so) the reaction is very fast has a high rate (1) reaction would be complete / most would have reacted in 30 seconds (1)	2			1 1		1 1
	(c)	(i)		frequency that is absorbed by one reactant only but not by its product (or vice versa) accept 'frequency absorbed by only one species'	1		1			1

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
		(ii)		gradient = -1.60×10^3 (allow range –1.57 to –1.63) (1) gradient = $\frac{-E_a}{R}$ (1) E_a = 13.3 kJ mol^{-1} (allow range 13.0-13.5) (1)	3		1	1 1	1 1 1	
					13	**4**	**5**	**4**	**3**	**3**

